

Metaphor analysis on pre-service early childhood teachers' conception of AI (Artificial Intelligence) education for young children

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ABSTRACT

This study explored PEC teacher' (pre-service early childhood teachers') conception of AI (Artificial Intelligence) education for young children by using metaphor analysis. Through exploring the conceptual metaphors of AI education for young children with 137 undergraduate students majoring in ECE (early childhood education) at I University in the central part of the United States, this present study tried to figure out the educationally meaningful direction of AI education for young children. The surveys were composed with open questions to find out PEC teachers' metaphor of AI education for young children and they were distributed to research subjects. Collected data were named, classified, categorised, and finally, 7 categories out of 137 metaphors were analysed through the step of confirming. PEC teachers in this study had positive metaphorical concepts such as 'possibility of play and experience', 'future essentials', 'innovation and change', 'convenience', 'assistant teacher', and negative metaphorical concepts such as 'double-sided meaning', 'Complexity' of AI education for young children. PEC teachers' beliefs about teaching-and-learning theory and methods related to AI education in ECE settings can be an important indicator of what thoughts and attitudes teachers will have when they implement the AI-related curriculum. Therefore, since PEC teachers have direct responsibility of planning and implementing AI education at ECE institutions, their conception of AI education in ECE would provide an important insight for shaping AI education for young children in the future.

1. Introduction

It dates to the 1940s that humans began studying AI (artificial intelligence) (Denning & Tedre, 2019). The attempt to make a machine that thinks like a human being through long research has been remarkably developed due to the recent advancement of big data and deep learning (Ali et al., 2021; Vartianinen, Tedre & Valtonen, 2020). Big data, clouding computer, and the Internet of Things (IoT), which are key technologies leading the era of the 4th industrial revolution, have developed AI technology to exert a tremendous influence on human daily life, which is bringing about major changes in human politics, economy, society, and culture as a whole (Zhang & Aslan, 2021). Such AI continues to develop with the ability to perceive, infer, understand natural languages, and learn as humans do (Ouyang & Jiao, 2021). Now, AI is very close to our lives, such as managing machines remotely through our smartphones, doing what we want by voice-commanding through Bixby or Siri, and being able to drive conveniently using autonomous vehicles.

However, AI also has harmful aspects and the ethical issues related to AI technologies should be considered promptly (Ouyang &

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Jiao, 2021). In regards of machine learning and deep learning, AI developers have to make sure to train AI machines with equal amounts of data sets without any bias embedded in AI models (Ali et al., 2021). AI still perceives white people as human beings by using facial-recognition programs while black people as materials or black box. AI responses male nurse practitioners as doctors while female doctors as nurses. The phenomenon AI perceives male or female and white or black differently is caused by inequitable data amounts human AI developers provide when training AI (Mercer, Hennessy & Warwick, 2019). The data transparency, unbiased machine learning, privacy and regulation on AI, and security of personal information would be more and more important in the future society in order to keep human beings from being isolated in AI-prevailed society in terms of the notion of digital divide (Frana & Klein, 2021). Therefore, for prospective students who will be future citizens in AI society, education is much important to include the advantage and disadvantage of AI technologies and how human beings can use AI for better well-beings and societal development (Vartiainen et al., 2020).

Due to the development of AI technology, AI education to cultivate future citizens is being included in the national curriculum (Ali et al., 2021). The United States is implementing supportive projects for AI education under the leadership of the National Science Foundation (NSF), and has established organisations such as AI4K12 and AI4All to develop five Big Ideas announced by AI4K12 (Zhang & Aslan, 2021). AI's five big ideas were presented in the guidelines for AI curriculum published by AI4K12 in the United States, which is run by NSF (National Science Foundation), AAAI (American Artificial Intelligence Association), and CSTA (Computer Science Teachers Association) (Ouyang & Jiao, 2021). In addition, many countries, including the UK, China, and Australia, are starting AI education in their countries, and some nations are in elementary school or higher, but most countries have AI education plans begun from preschool (Bardakci & Unver, 2020). The curriculum development in national level attempts to include AI education in K-12 curriculum, but still in preschools and kindergartens in USA, AI education is not prevailed because ECE educators think that digital technology in education is not suitable for young children's development (Zhang & Aslan, 2021). However, AI4K12 aims to educate the future citizens for AI concepts and usages including ethical values related to AI. In order to balance between everyday practice and governmental policy, educators should consider the societal changes and needs which influence on educational practices (Keengwe & Onchwari, 2009; Kim et al., 2013). In particular, the acceleration of the COVID-19 pandemic into a digital society has made AI education more interested and important in the field of education (Ali et al., 2021).

The role of teachers is very important for AI education to be implemented in educationally meaningful ways in everyday classroom situations (Mishra & Koehler, 2006; Parasuraman, 2000; Shulager & Fusco, 2003; Wegerif, 1996; Virtanen et al., 2020). Teachers are important beings in planning, implementing, and evaluating teaching-and-learning, and teachers' beliefs and attitudes are major factors that determine both the content and method of classroom activities (Koehler & Mishra, 2009; Mercer et al., 2019; Kim et al., 2013). Teachers who value the need for science education on environmental issues caused by global warming tend to treat the scientific contents more importantly in their classrooms. Teachers with beliefs that emphasise on the importance of creativity or personal character education always pursue this direction in their education (Cullen & Greencem, 2011; Fang, Chung & Yun, 2022; Kang & Kim, 2017). Therefore, for achieving the successful application of AI education in educational field, it is very important how teachers who conduct AI education in their educational fields think of AI education and recognise its importance and necessity (Avidov-Ungar, Shamir-Inbal & Blau, 2020; Ouyang & Jiao, 2021; Zhang & Aslan, 2021). Teachers' perception and beliefs shape their pedagogical practices and influence on their interaction with young children in early childhood classroom settings (Ali et al., 2021; Avidov-Ungar et al., 2020). When teacher has a positive belief and attitude on AI education, the educational practice might be much more active and responsible (Kim et al., 2013). However, when teacher's attitude and perception are negative on AI education, (s)he would be reluctant to apply AI education in his (her) classroom (Langub & Lokey-Vega, 2017; Lim, 2015). Accordingly, exploring teachers' perception on AI education in ECE might be an index of the possibility of successful AI education in ECE settings.

As a matter of fact, in the field of education, various factors affect and shape the educational solution and pedagogical practices (Cullen & Greencem, 2011; Lim, 2015). Teacher's attitude and belief influence the successful performance of educational pedagogy, but there are more circumstances educators should consider when we consider the improvement of education (Kim et al., 2013). To apply AI education into every classroom setting, pedagogical contents and curriculum development related to technology integration in ECE should be dealt with and supported in educationally meaningful ways (Blau, Grinberg & Shamir-Inbal, 2018). Technological content knowledge should be embedded in teacher professional training and continuing education for teachers (Koehler & Mishra, 2009). Identifying the merits and also the shortages in AI technologies, and unbiased approach for teaching and learning should be considered at first before applying AI technologies with young children in ECE (Langub & Lokey-Vega, 2017). With setting up all these necessary elements for improving AI education in practices, teacher's perception and teaching beliefs are prioritised for completing the authentic AI educational implements with young children in developmentally appropriate ways (Ali et al., 2021; Avidov-Ungar et al., 2020).

To find out human's perception on certain concept, metaphors have been used in research for a long time. By analysing metaphors which humans think of certain objects or phenomena, metaphors help us understand abstract and unfamiliar concepts as concrete and familiar concepts by utilising the linguistic framework that exists within our thoughts (Amin, 2015; Guerrero & Villamil, 2002; Martinez, Saulea & Huber, 2001). In education, metaphors have been used in studies to find out teachers' beliefs, values, and attitudes toward education (Tobin, 1990). In addition, metaphors have also been utilised in classroom activities for helping learners understand certain complex concepts (Zhao, Coombs & Zhou, 2010). As such, metaphor analysis has been used in various topics in the field of pedagogy, and metaphor analysis can be used to find out the teachers' conceptual system and belief system. Looking at previous studies related to metaphors in education so far, there have been studies on teachers' leadership-related policies, the concept of happiness felt by early childhood teachers, and the perception of educational concepts of play, etc. (Amin, 2015; Avidov-Ungar et al., 2020; Blau et al., 2018; Koc, 2013; Saban, Kocbeker & Saban, 2006). For PEC teachers, there were studies on preschool teachers' perception about the policy of day care centre, and the personal characteristics of early childhood teachers (Cullen & Greencem, 2011;

Pinnegar, Mangelson, Reed & Groves, 2011).

This study aims to find out how PEC teachers, who would play an important role in ECE in the future, recognise and define AI education for young children. The results of this study would serve as basic data for developing educational programs to enrich AI education-related competencies of PEC teachers. For this purpose, the research questions of this study are as follows.

- (1) What are the metaphorical expressions of PEC teachers on AI education for young children?
- (2) What are the characteristics of types of metaphor of PEC teachers about AI education for young children?

2. Literature review

2.1. AI (Artificial Intelligence) education

Artificial intelligence generally refers to technology that allows machines to have intelligence to think and process like humans (Vartiainen et al., 2020). In other words, it allows machines to recognise, think, perform, and solve problems like humans (Denning & Tedre, 2019). AI is a sub-area of computer science that can think logically, enables machine learning and understanding natural languages, and includes a wide range of technologies such as computer and robotics (Ouyang & Jiao, 2021).

AI technology first began to be studied in the 1950s when the AI concept was established by John McCarthy and others but has experienced dark ages several times due to technical limitations (Zhang & Aslan, 2021). However, due to recent remarkable advances in computer performance and rapid advances in algorithms, rapid growth has been achieved in various fields of society (Ali et al., 2021). It has become a national task to cultivate AI-related capabilities for learners who are future citizens around the world by leaping forward as a key technology that will lead the era of the 4th industrial revolution (Avidov-Ungar et al., 2020). Recently, the world is trying to take the lead in the future industry by including AI education in the regular curriculum to educate and cultivate AI-related citizens.

As such, competition for AI is intensifying around the world, and the speed and scope of social changes that AI will bring to mankind is expected to become faster and wider (Ouyang & Jiao, 2021). Led by advanced countries, it is establishing a national strategy of AI which includes a national vision to prepare for the era of AI, and its main contents are to foster talented students in the field of AI and establish an AI education system for the whole country (Vartiainen et al., 2020). Not only software education but also AI education is expanding in the field of education, and there are increasing voices about the need for embracing related contents in pre-service teacher education programs (Mercer et al., 2019).

In the field of education, research on AI education is being actively conducted in two aspects: one is for 'learning about AI', and the other for 'learning with AI' (Zhang & Aslan, 2021). First, 'learning about AI' refers to learning to understand the concepts and components of AI technologies such as computer science, natural language processing, computer programming & coding, and machine learning in the field of AI (Mercer et al., 2019). In other words, it consists of understanding the important core concepts of AI, dealing with the scope and machine learning of AI technology, the human-computer interaction related to AI, and the social impact of AI. To learn about AI, the leading countries in the field of AI such as the United States, the United Kingdom, and China operate AI education programs through online learning platforms (Ouyang & Jiao, 2021). Examples of representative AI education programs aimed at introducing AI education in K-12 education include online platforms such as Ready AI in the United States, Machine Learning for Kids in the United Kingdom, and Google's Teachable Machine, and the world's first AI textbook developed in 2018 (Frana & Klein, 2021).

Next, 'learning with AI' refers to the use of AI in education as a tool for learning (Zhang & Aslan, 2021). In terms of AI education in the field of ECE, it means a direction in which young children can go through various experiences and broaden their thoughts while playing with AI-embedded programs. It also includes allowing young children's play to be flexible and diverse through interaction with AI agents such as AI robots, AI computers, or AI devices (Vartiainen et al., 2020). This is related to global IT companies such as Google, Facebook, Microsoft, and IBM investing heavily in developing various software that can increase the possibility of individualised learning using AI technology or assist teachers in education and teachers' administrative works (Frana & Klein, 2021). With the recent activation of online virtual education due to COVID-19, the use of AI technology in education is being actively studied and tested for its potential. Although research related to the movement to introduce AI into education is still expanding quantitatively, substantial research on what AI education should aim for and what should be desirable direction of AI education is still scarce.

2.2. Metaphor analysis

Cognitive linguists said that investigating the expression of concept that humans have about a specific object is a way of recognising the world around them, at the same time, it is a way of understanding types of human behaviours, and a way of establishing relationships with others (Schmitt, 2005). This conceptual system plays an important role in defining objects faced by individuals in the real world (Amin, 2015; Shaw, Parsons & Vasinda, 2021). Since human language is expressed based on an individual's thinking, behavioural style, and emotions, it can be said that the general conceptual system which an individual has in mind for a certain object appears through the everyday language expression of metaphor (Goldstein, 2005; Koc, 2013).

Shaw et al. (2021) argued that metaphors are not just literary grammar or skills used in literature arts, but everyday skills of expressing human beings themselves. Metaphoric expression that individuals use in their lives is systematically linked to a mindset called conceptual metaphor (Bozlk, 2002). In other words, the conceptual metaphor used by humans is formed from a thinking system based on individual experiences and is expressed as a metaphorical language expression for a specific object. The reason why many educational studies have used metaphor analysis so far is that they can provide a variety of information such as teachers' thoughts on

specific subjects, perceptions of curriculum, attitudes, and values for teaching-and-learning (Johnson, 2001; Martinez et al., 2001; Saban et al., 2006). The metaphor for a teacher's specific object indicates how the teacher perceives and interprets the curriculum, which has an important meaning indicating teacher's disposition of pedagogical practices (Zhao et al., 2010).

Metaphor analysis broadened humans thinking system of certain phenomena and interpretation toward certain situations (Pinnegar et al., 2011). Especially, when understanding more abstract concepts of person's perception, metaphor provides familiar conceptual tools to express their thoughts and representation in the linguistic framework. (Amin, 2015; Guerrero & Villamil, 2002; Martinez et al., 2001). In educational research, metaphors have been used to find out teachers' beliefs systems (Avidov-Ungar et al., 2020; Blau et al., 2018; Koc, 2013). Teacher's values and attitudes toward education would be reflected on metaphorical analogy (Tobin, 1990). Besides exploring teacher's perspective, metaphors have also been used with children in classroom situations for learning and help learners understand complex concepts by using metaphor strategy in teaching-and-learning practices (Saban et al., 2006; Zhao et al., 2010).

Since the metaphor is based on teacher's experience as the subject of the curriculum, it can provide various information on the propensity of future curriculum implementation at the same time as the inclination of curriculum improvement (Saban et al., 2006; Shaw et al., 2021). Metaphor analysis as a research method is a method of analysing the concept, system of understanding, or social meaning of an individual or group through interpretation of an individual's metaphorical expression (Shaw et al., 2021). The procedure for metaphor analysis presented by Saban et al. (2006) and Schmitt (2005) is (1) the naming and labelling stage, (2) the classification stage, (3) the categorisation stage, (4) the reliability evaluation stage, and (5) the quantitative analysis stage. In this metaphor analysis study, it is important to interpret and categorise the meaning of metaphors and secure reliability while constructing the metaphorical categorisation.

3. Research method

3.1. Subjects

A survey study was conducted on 137 undergraduate students majoring in early childhood education at I University in the central part of the United States from October 18 through November 19, 2021. PEC teachers' prior experience of taking AI-related courses were asked before collecting research consent forms. 65 participants replied they had experience of taking AI-related courses, but none of participants took any courses related AI education in ECE. The general characteristics of research participants are shown in Table 1.

145 questionnaires were originally distributed, but finally, 137 responses were used for data analysis, excluding those that were not metaphorical or could not be metaphorically analysed. The students who participated in the study were 36 freshmen (26.4 %), 31 sophomores (22.6 %), 35 juniors (25.5 %), and 35 seniors (25.5 %), with 16 male students (11.7 %) and 121 female students (88.3 %) among them. Their race were 48 Caucasians (35.0 %), 43 African Americans (31.4 %), 28 Asians (20.4 %), 9 American Indians (6.6 %), 6 Pacific Islanders (4.4 %), 1 Latinos (0.7 %), and 2 others (1.5 %). In addition, to the questions about their experience of taking AI-related courses, 65 participants answered "yes" (47.4 %) and 72 participants replied "no" (52.6 %), and all of 65 respondents took courses related AI in general, not AI education.

3.2. Measurement tool

In order to find out PEC teachers' metaphor of AI education for young children, systematic metaphor analysis used by Schmitt (2005) and Saban (2006) was utilised based on the study of cognitive linguistics-based metaphors by Shaw et al. (2021). Systematic metaphor analysis is a method of analysing and exploring the conception about a specific topic by describing the linguistic metaphor and explaining the reason why the metaphor was expressed.

Table 1
General characteristics of the research subjects ($N = 137$).

		Number of subject (N)	Ratio of subject (%)
Grade	Freshman	36	26.4
	Sophomore	31	22.6
	Junior	35	25.5
	Senior	35	25.5
Gender	Male	16	11.7
	Female	121	88.3
Race	White/Caucasian	48	35.0
	African American	43	31.4
	Asian	28	20.4
	American Indian	9	6.6
	Pacific Islander	6	4.4
	Hispanic/Latino	1	0.7
	Others	2	1.5
	AI course experience		
	Yes	65	47.4
	No	72	52.6
Total		137	100

Two questions were asked to analyse the conceptual metaphor of PEC teachers about AI education for young children. An open-ended sentence-completion questionnaire was used. Question 1 asked to complete the blank with describing the metaphor of AI education through the question "AI (Artificial Intelligence) education for young children is ()." Question 2 asked to describe the reason by filling out the blank in detail through the question "Because ()." to find out the meaning of the metaphor expressed in AI education in question 1. The questions of the research tool are shown in Table 2.

3.3. Research procedure

First, a preliminary survey was conducted from October 6 through October 12, 2021, and the same questionnaire was used for 10 PEC teachers enrolled in the department of ECE at I University. Based on the results of the preliminary survey, a guide phrase was inserted to help PEC teachers understand the meaning of the term of 'AI education for young children', and as a result, all questions were confirmed to be used for the present study because all the respondents replied accordingly with metaphor and reason without any difficulty.

Before the survey for this present study, the researcher guided the purpose and method of this research to all participants, and explained the meaning of AI and what AI education in ECE means and checked out the participant's understanding. The researcher obtained prior consent to participate in the study, and conducted this survey with 145 PEC teachers who signed the consent form of research participation in the study. This survey was conducted on 145 PEC teachers majoring in ECE at I University for four weeks from October 18 through November 12, 2021, and only 137 of the collected questionnaires were finally analysed, excluding 8 that were not metaphorical or could not be metaphorically analysed.

After analysing the collected data, to increase the reliability of qualitative data analysis, three ECE experts analysed 30 randomly selected questionnaires, and then discussed and adjusted the meaning and categorisation of questionnaires with different opinions compared to the researcher's original analysis to confirm the final categorisation of metaphor for data analysis.

3.4. Data analysis

The metaphor analysis of the collected data was conducted according to the systematic metaphor analysis procedure of Saban et al. (2006) and Schmitt (2005), and the metaphor analysis step as follows. First, in the naming and labelling stage, the metaphorical expressions of EC-AI education of the study participants were numbered from 1 to 137, and 'metaphor' and 'the reason of the metaphor' were coded into the Excel program. Second, in the classification stage, the researcher nomadised the first stage of data focusing on similarities and commonalities among metaphors, and 125 metaphors were derived by combining some similar metaphors together. Then the researcher randomly selected 20 questionnaires and reviewed with three ECE experts to verify the researcher's classification. All opinions were agreed to complete the classification according to the researcher's classification. Third, based on the commonalities and characteristics between the metaphors and reasons derived in step 2, the category name was determined to describe each category, and to express the meaning of the category properly. In this process, a total of 8 categories were derived: 'possibility of play and experience', 'future essentials', 'innovation and change', 'difficulties', 'double-sided meaning', 'convenience', 'assistant teacher', and 'complexity', and set within the category. Later, it was found that there was ambiguity in the category in the process of continuously reading with an emphasis on the similarity of the system and items of content and the normalisation of ambiguous parts. Through the meeting, the category was reaffirmed, and the category of 'difficulty' was re-categorised into a sub-category of 'complexity' and finally decided into 7 categories. Since then, a total of 7 categories were finally agreed. Fourth, a review of three ECE experts was conducted on the researcher's final categorisation one more time. Finally, the results of the categorised metaphor classification and the reasons for the metaphor were reviewed, and the frequency analysis of the final categorisation results was conducted using SPSS 20.0 for windows. The specific procedure of metaphor analysis is shown in Fig. 1.

Table 2
Questions of the research tool.

Question Number	Questions
1	AI (Artificial Intelligence) education for young children* is _____.
2	Because _____ _____ _____.

*AI (Artificial Intelligence) education for young children: This means an education that allows young children to understand AI, form the right values, and express their thoughts and feelings freely through play through AI while playing and interacting with AI.

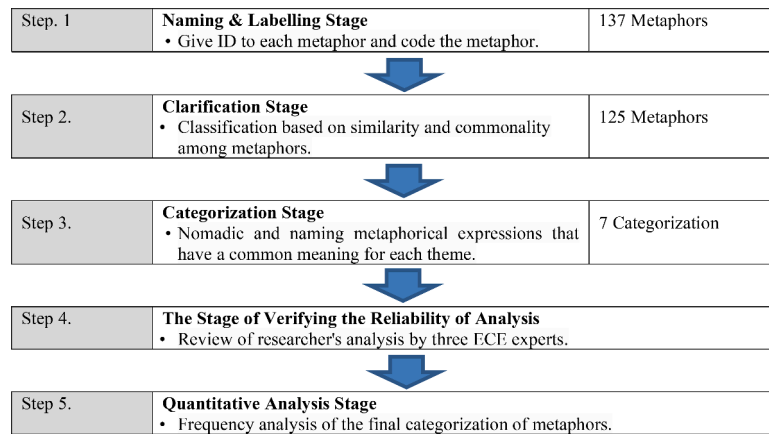


Fig. 1. Process of metaphor analysis.

4. Results

4.1. Results of categorisation of metaphors about AI education for young children

To analyse the results of metaphorical responses about AI education for young children by 137 research participants, types of metaphors were categorised into 7 categorisations: 'possibility of play and experience', 'future essentials', 'innovation and change', 'double-sided meaning', 'convenience', 'assistant teacher', and 'complexity'. In the process of categorising the metaphors about AI education for young children, the categorisation was determined by referring to the description of the reasons for the metaphor, not only the words or phrases answered.

PEC teachers' metaphor about AI education for young children could be categorised into possibility of play and experience (24.2 %), future essentials (22.6 %), innovation and change (16.8 %), double-sided meaning (13.1 %), convenience (9.5 %), assistant teacher (8.0 %), and complexity (5.8 %). In the 'possibility of play and experience' classified as the most metaphor type, metaphors such as hope, road, key, baby, white drawing paper, and seed appeared in overlapping forms. Next, in the 'future essentials', metaphors such as air, water, meal, essential subject, safety helmet, and smartphone were identified as overlapping metaphors. In addition, meeting, compass, and chameleon were duplicated in the 'innovation and change', and both sides of coin, double-sided colored paper were duplicated in the 'double-sided meaning'. Electric vehicle, wheel, and robot were duplicated in the 'convenience', while the 'assistant teacher' was helper, responsibility, and the 'complexity' showed metaphorical forms such as space and maze without overlapping metaphors. These types of metaphors of PEC teachers reflect the structure of thinking about AI education for young children that they usually think of and have inside their mind. Table 3 shows the results of categorisation of metaphors about AI education for young children by PEC teachers.

Table 3
Categorisation of metaphors about AI education for young children ($N = 137$).

Categorisation	Metaphor (N. of Response)	Number of response (%)
Possibility of play & experience	Hope (3), Possibility (2), Road (2), Key (3), Baby (2), Chick (1), Seed (3), First love (1), White drawing paper (3), Colored paper (1), Fire (2), Joy (1), Heart (1), Mind (1), Snack (1), Start (1), Picnic (1), High jump (1), Starting line (1), Running away (1), Opportunity (1)	33 (24.2)
Future essentials	Air (10), Water (5), Light (1), Essential subject (2), Artificial respiration (1), Safety helmet (2), Smartphone (2), Money (1), Navigation (1), Meal (3), Life (1), Breathing (1), Essential Vitamin (1)	31 (22.6)
Innovation & change	Meeting (4), Compass (3), Innovation (1), Navigation (1), Shield (1), Rainbow (1), Peacock (1), Prism (1), Neon Sign (1), Chameleon (4), Wind (1), Revolution (1), Relationship (1), Mentor (1), Challenge (1)	23 (16.8)
Double-sided meaning	Both sides of coin (3), Double-sided colored paper (2), Innovation (1), Double-edged sword (1), Revolution (1), Sugar (1), Future (1), Smartphone (2), Mobius strip (1), Life (1), Industrial revolution (1), Half-watered cup (1), Sunlight (1), Horizon (1)	18 (13.1)
Convenience	Electric vehicle (2), Wheel (2), Tool & machine (1), Vending machine (1), Shortcut (1), Robot (2), Washing machine (1), Car (1), Airplane (1), Bed (1)	13 (9.5)
Assistant teacher	Helper (2), Assistant (1), Mascot (1), Playmate (1), Children's book (1), Assistant teacher (1), Teacher (1), Manager (1), Responsibility (2)	11 (8.0)
Complexity	Space (1), Maze (1), Endless sea (1), Compass-free voyage (1), Dark night sky (1), Question mark (1), Beginner driving (1), Tangled thread (1)	8 (5.8)
Total	137 (100 %)	

4.2. Characteristics of metaphors about AI education for young children

The characteristics of each 7 metaphors derived through metaphor analysis on AI education for young children by PEC teachers are as follows.

4.2.1. Metaphor of 'possibility of play and experience'

As a result of analysing metaphors for EC-AI (early childhood artificial intelligence) education, it was found that 33 (24.2 %) PEC teachers perceived EC-AI education as 'possibility of play and experience.' Examples of metaphors in 'possibility of play and experience' for EC-AI education include 'Hope,' 'Key,' 'Seed,' and 'White Drawing Paper.' In accordance with the current situation in which AI education is not in full swing, the possibility of play and experience of young children using AI was presented as a metaphor. In the case of seeds, it seems invisible now, but the meaning of having infinite possibilities for something big and wonderful to happen in the future is described as the reason of metaphor. Table 4 shows metaphorical expressions and examples of reasons in 'possibility of play and experience.'

4.2.2. Metaphor of 'future essentials'

As a result of analysing metaphor for EC-AI education, 31 PEC teachers (22.6 %) recognised EC-AI education as 'future essentials.' 'Air,' 'water,' and 'meal' were identified as types of metaphors in the 'future essentials' area about AI education for young children. In the sense that AI education is an inevitable educational trend in the era of the 4th industrial revolution, indispensable factors for maintaining human life such as air, water, and meal appear as metaphorical expressions. In addition, it also implies that future AI education would be a prevailing educational method and content like air and water in human lives. Table 5 shows examples of metaphorical expressions and reasons in the 'future essentials' area.

4.2.3. Metaphor of 'innovation and change'

As a result of analysing metaphor about AI education for young children, 23 pre-service teachers (16.8 %) perceived EC-AI education as 'innovation and change.' Types of metaphors in 'innovation and change' for EC-AI education include 'meeting,' 'compass,' and 'chameleon.' This type of metaphor means that AI education would play an important role in innovation and change in the direction of future education and includes expectations that innovative education reform and development which have not been encountered so far would be achieved through AI education. Table 6 shows the metaphorical expression and reasons in 'innovation and change.'

4.2.4. Metaphor of 'double-sided meaning'

As a result of analysing metaphor about AI education for young children, it was found that 18 (13.1 %) pre-service teachers perceived EC-AI education as a 'double-sided meaning.' Types of metaphors in the 'double-sided meaning' area for EC-AI education include 'both sides of coin,' 'double-sided colored paper,' and 'smartphone.' This type of metaphor can be seen as a position to consider not only the convenience and positive aspects of humanity that technological development in the AI era will bring, but also negative aspects such as risks and inequality that can be caused by AI technology. In the field of education, there will be aspects of positivity and negativity, so it means that ethical education of AI should be provided to young children. Table 7 shows examples of metaphorical expressions and reasons in the 'double-sided meaning' area.

4.2.5. Metaphor of 'convenience'

As a result of analysing metaphor about AI education for young children, it was found that 13 (9.5 %) PEC teachers perceived EC-AI education as 'convenience.' Types of metaphors in the area of 'convenience' for EC-AI education include 'electric vehicle,' 'wheel,' and 'robot.' This type of metaphor can be seen as reflecting the expectation that AI education would make future life more convenient and the perception that it will serve as an educational tool to make education easier and more comfortable than the past. Table 8 shows examples of metaphorical expressions and reasons in the 'convenience' area.

4.2.6. Metaphor of 'assistant teacher'

As a result of analysing metaphor about AI education for young children, it was found that 11 (8.0 %) PEC teachers perceived EC-AI education as an 'assistant teacher.' The types of metaphors of 'assistant teacher' for EC-AI education include 'helper' and 'responsibility.' This type of metaphor contains the expectation that AI education can be used as an auxiliary means to help early

Table 4
The metaphor and reason of 'possibility of play and experience'.

Metaphor	ID	Reason
Hope	101	Because it makes us look forward to the possibility of having a different thinking, different styles of play, and a different educational experience.
Key	74	Because I don't know what's going to happen, but it opens the door to something richer and greater than now.
Seed	23	Because it seems invisible now, but it has infinite possibilities for something big and wonderful to happen in the future.
White drawing paper	59	Because nothing is drawn and determined, so young children can draw any pictures as much as they want.

Table 5

The metaphor and reason of ‘future essentials’.

Metaphor	ID	Reason
Air	36	Because AI education is an inevitable direction of education in the future, like the indispensable air for all living things.
Water	112	Because AI education in the future is a must-do education in ECE, and it is essential, not an option, and therefore, it must be learned as a teacher.
Meal	62	Because it is a mandatory education for future citizens.

Table 6

The metaphor and reason of ‘innovation and change’.

Metaphor	ID	Reason
Meeting	106	Because the direction of future education brings about a new encounter that innovates in a very different direction than now.
Compass	66	Because it will lead to new educational innovations and developments that have not been encountered so far.
Chameleon	124	Because it will play a big role in bringing about new changes and innovations that are hard for us to predict.

Table 7

The metaphor and reason of ‘double-sided meaning’.

Metaphor	ID	Reason
Both sides of coin	107	Because not only the positive aspects of mankind that technological advances in the era of AI will bring, but also negative aspects such as risks and inequality that may arise must be considered.
Double-sided colored paper	132	Because when teaching AI to young children, value education related to good and bad things, and safety education to protect themselves must be provided together.
Smartphone	47	Because if you use it well, it is very convenient and good, but if you use it incorrectly, it can have an adverse effect.

Table 8

The metaphor and reason of ‘convenience’.

Metaphor	ID	Reason
Electric vehicle	9	Because good AI education will make future life more convenient.
Wheel	39	Because AI education is a tool that makes the educational environment easier and more comfortable than in the past.
Robot	86	Because you will be able to give a lot of ideas to replace what you’ve been struggling with.

childhood teachers’ education and administrative work. It can be seen as a perception that reflects interest in the use of AI technology to maximise educational interaction with young children. [Table 9](#) shows examples of metaphorical expressions and reasons in the ‘assistant teacher’ area.

4.2.7. Metaphor of ‘complexity’

As a result of analysing metaphor about AI education for young children, it was found that 8 PEC teachers (5.8 %) perceived EC-AI education as ‘complexity.’ Types of metaphors in the ‘complexity’ area of AI education for young children include ‘space,’ ‘maze,’ and ‘endless sea.’ This type of metaphor means that the concept of AI education itself is difficult and contains a sense of uncertainty about how to apply it specifically in education. In the future, it is necessary to consider the perception of AI education as an unpredictable educational environment that teachers have as a reason for the need to have AI literacy for EC-AI education. [Table 10](#) shows examples of metaphorical expressions and reasons in the ‘complexity’ area.

5. Discussion and conclusions

In the era of the 4th industrial revolution, AI education is included in the regular curriculum worldwide and future education for digital citizenship is being implemented ([Frana & Klein, 2021](#)). AI technology will continue to develop in the future society, so more understanding and knowledge of AI will be needed than ever ([Vartiainen et al., 2020](#)). Globally, the students who need to participate in AI education ranges from pre-schoolers to adults, and discussions and research on the content and methods at each school age are

Table 9

The metaphor and reason of ‘assistant teacher’.

Metaphor	ID	Reason
Helper	46	Because teachers will be able to focus on education because they can help with tasks such as calling attendance and reading fairy tales.
Responsibility	55	Because the teacher plays the same role as an assistant teacher in the classroom so that the teacher can secure more time to interact with infants.

Table 10

The metaphor and reason of ‘complexity’.

Metaphor	ID	Reason
Space	98	Because it is such a broad and difficult concept that I don't know where to start and how to start.
Maze	130	Because it feels unpredictably difficult, so I don't know what to do with infants.
Endless sea	11	Because it is so difficult and feels far away, so it is difficult to know how to incorporate it into education.

continuously needed (Denning & Tedre, 2019; Ouyang & Jiao, 2021). When AI education is conducted in ECE, the competence of the teacher in charge of the education in classroom should be cultivated first (Ali et al., 2021). For EC-AI education in the future society, it is necessary to cultivate the knowledge of AI education of PEC teachers (Keengwe & Onchwari, 2009; Plowman, McPake & Stephen, 2012). To do so, there are various factors must be considered in curriculum development, such as pedagogical content knowledge, educational environments building, and teacher professional development programs for continuing training for teacher's digital competence, etc. (Cullen & Greencem, 2011; Lim, 2015). Among them, this study focuses on teacher's belief and perception on educational practices. Therefore, this study aims to find out the PEC teachers' perception of AI education for young children through the research method of metaphor analysis.

First, PEC teachers had positive perspectives about AI education for young children even though AI is still difficult concept for teachers to master. PEC teachers' positive metaphorical concepts were such as ‘possibility of play and experience,’ ‘future essentials,’ ‘innovation and change,’ ‘convenience,’ and ‘assistant teacher.’ When recognised by PEC teachers, AI education in ECE is also an inevitable element in the future society and has the potential to play an important role in expanding and deepening young children's play and their educational experiences. Young children also had prior experience on AI technologies in everyday lives and they saw AI not only as to learn, but also as to play with (Vartiainen et al., 2020). Moreover, when AI technologies develop rapidly and support everyday classroom practices as assistive teaching agents, teachers would be able to invest more time to interact with young children and concentrate on educational practices and teachers could focus more on qualitative interaction with young children in play-centred atmosphere (Lim, 2015; Liu, 2012; Vygotsky, 1978). Since teachers majored in ECE which is not related to computer science, the concept of AI itself was not familiar to them, and also, PEC teachers recognised that they were not prepared to apply AI education with very young children in early childhood classroom settings. Considering the developmental stages of young children, the importance of AI education in terms of the ethical issues related to AI should be emphasised (Vartiainen et al., 2020).

Second, based on the emerging categories from the metaphors for EC-AI education, most participants positively perceived EC-AI education as possibility of play and experience (24.2 %), future essentials (22.6 %), innovation and change (16.8 %), convenience (9.5 %) and assistant teacher (8.0 %). According to the research by Mercer et al. (2019) and Avidov-Ungar et al. (2020), ECE teachers perceive AI education in timely and innovative instructional paradigm which educators should follow up and try to learn for applying to their classroom settings. AI education should be performed with play-based and child-centered experience in ECE, and the play-oriented ways need to be recognised as desirable direction of EC-AI education (Ouyang & Jiao, 2021; Vartiainen et al., 2020; Zhang & Aslan, 2021). In addition, EC-AI education can be an important medium for changing toward innovation and development of education in the future (Ali et al., 2021).

Third, PEC teachers are also negatively aware of EC-AI education. The double-sided meaning (13.1 %) and complexity (5.8 %), that is, a total of 18.9 % among responses, awakens the negative aspects that AI might bring to human society, suggesting that the ethical issues and values related to AI technologies must be dealt with in AI education. According to a study by Langub and Lokey-Vaga (2017), digital literacy education must also deal with cyber safety and security in addition to the positive aspects of digital technology in order to help future citizens construct educational values for using digital technologies in desirable and educationally meaningful ways. In AI education for young children, ethical education should be of paramount importance to prevent young children from the dangers in cyber space in AI society and to support them to learn how to create a beautiful world where AI and humans collaborate together (Ali et al., 2021; Mercer et al., 2019; Vartiainen et al., 2020). According to Frana and Klein (2021), ‘education of understanding AI’ and ‘education of utilising AI’ are not exclusive each other, but overlap with ‘education of ethical issues on AI,’ and AI education should put a top priority of introducing the ethical aspects of AI to children in order to prepare for them to be sensitive to ethical issues and safety concerns in digital world.

The limitations of this study are as follows. First, this study was explored through metaphor analysis to find out PEC teachers' perception about AI education for young children. In metaphor analysis, there is an advantage of seeing the teacher's thinking system, but it is difficult to explore the actual aspects and contents of perception and demand, so research on perception and needs is also needed in the future. Second, this study has limitations in generalising to all PEC teachers due to the size of sampling, the male teacher to female teacher ratio, or the limitations of sampling method, so further research on various research subjects needs to be continued. The imbalance of male to female PEC teachers are due to the national ratio of male-to-female teachers in the US. Third, research on the perception of PEC teachers is needed to develop high-quality PEC teachers' education programs, but more in-depth research on educational goals, contents, methods, and evaluation should be actively conducted through practical study, qualitative research, delphi study, and literature review. Fourth, in the present study, each participant was allowed to answer only one metaphor at a time. It might set a boundary of selecting only one aspect of AI-related perception even though they might have any contrasting or contradicting beliefs on AI because AI is neither “all good” nor “all bad.” In the future research, allowing each respondent to have both a primary and secondary metaphor to describe their perception and beliefs on AI would be more worthwhile to express their deeper insights on it.

Based on the above conclusion, the researcher would like to make the following suggestions. First, in the future AI curriculum for

PEC teachers, education about ethical issues related to AI and the social impact of AI needs to be included as content elements of professional development programs. In order for prospective teachers to perceive AI in balanced views, the advantages and disadvantages AI might have should be dealt with young children in education. Therefore, AI education contribute to cultivating the future citizenship for young children. Second, based on the positive perception of AI by PEC teachers, AI literacy education should be included to help young children expand their play and experience by utilising AI programs in play-centred ways. For the development of a future society where humans and machines can collaborate, it is necessary to develop PEC teachers education programs that cultivate AI literacy as core competence. To successfully establish AI education through regular curriculum in the future, it is necessary to develop a curriculum that considers social and cultural contexts based on the thinking system of PEC teachers who will directly interact with young children in classrooms and be in charge of AI education for young children in ECE settings.

Ethics approval statement

All subjects participated voluntarily after signing their written informed consent forms for participating in this study. The participants were informed that they would be possible to withdraw from this study anytime they want.

CRediT authorship contribution statement

Eun Mee Lim: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Software, Supervision, Validation, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Data availability

The data that has been used is confidential.

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